
INTERNSHIP MATCHING DATABASE FOR BUS-395 STUDENTS

FINAL PROJECT PROPOSAL

ITM 354

Group 6

Professor Eduard Merc

University of Hawai'i at Mānoa

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EXECUTIVE SUMMARY

Every BUS 395 student at the Shidler College of Business is required to land an internship, yet the information they need to do that is scattered everywhere. Listings arrive through email, Handshake, professors, advisors, and company contacts, with no single place to find them and no easy way to tell which ones fit a student's major and skills. Students lose opportunities due to missed deadlines, and advisors spend hours chasing down where each student stands.

We propose the Internship Matching Database, a single system that pulls all this together. It gives students one place to discover internships matched to their major and skills, and it gives advisors a live view of who has applied, who is interviewing, and who has received an offer. The result is less time spent searching and tracking, better matches between students and employers, and stronger outcomes that the program can measure. The pages that follow lay out the problem in detail, the solution we will build, how it is designed, and the value it delivers.

THE PROBLEM

The Shidler College of Business requires BUS 395 students to complete an internship before they graduate. That requirement is sound, but the process around it is not. Today, finding an internship depends on luck and legwork. A Management Information Systems student might hear about one opportunity from a professor, while a Finance student stumbles onto a completely different listing on Handshake. Neither has a full picture of what is available, and neither can quickly tell which openings match their major, their skills, or their timeline.

This creates real costs. Students apply late or miss deadlines entirely because the information reached them too slowly. They waste effort on roles they are not qualified for, and they overlook strong

matches simply because no one connected the dots. Strong candidates slip through the cracks, and the quality of the matches the program produces suffers.

Advisors and faculty feel the strain just as much. With no central record, they track student progress one conversation at a time, asking each student whether they have applied, interviewed, or received an offer. There is no reliable way to see the whole cohort briefly, which makes it hard to step in when a student is falling behind and harder still to report on how the program is performing.

THE OPPORTUNITY

The good news is that this is a solvable problem, and solving it pays off quickly. The core issue is not a shortage of internships; it is that the information about them is fragmented. Bring that information into one place, organize it by major and required skills, and connect it to a simple way to apply and track progress, and most of the friction disappears.

A centralized system turns scattered, out-of-date listings into a single, trustworthy source that students can search in seconds. It replaces guesswork with relevance by surfacing the internships that fit each student. And it gives advisors the real-time visibility they need to support students proactively and to show the program's results with hard numbers. The value shows up as time saved, better matches, fewer missed opportunities, and a program that can demonstrate its impact.

OUR SOLUTION

We will build an Internship Matching Database, a centralized system designed specifically for BUS 395 students at the Shidler College of Business. In one place, it brings together student profiles, majors, skills, internship listings, companies, application deadlines, and application status. From that

foundation, it does the two things the program needs most: it matches students to the right internships, and it gives advisors a clear view of every student's progress.

For students, the system becomes the first and only place they need to look. It shows internships matched to their major and skills, lets them search and filter by company, deadline, and keyword, and lets them apply in a single step. For advisors and faculty, it becomes a command center. They can post and maintain listings, see exactly where each advised student stands, and pull reports that summarize applications, interviews, and offers across the whole cohort. The sections below describe what the system does and how it is built to be dependable.

WHAT THE SYSTEM DOES

The capabilities below are the heart of the solution. Each one targets a specific piece of the problem and delivers a specific benefit to students, advisors, or both.

ONE PROFILE THAT DRIVES BETTER MATCHES

Each student maintains a single profile with their contact information, major, GPA, expected graduation date, and skills. This profile is what powers the matching, so students get suggestions that fit them rather than a flood of unrelated listings. Advisors have their own profiles and are connected to the students they support.

A LIVING, ORGANIZED CATALOG OF INTERNSHIPS

Faculty and advisors post and maintain internship listings, each one recording the title, description, hosting company, the major it targets, the skills it requires, the application deadline, pay rate, and location. Because every listing is tied to a major and a set of skills, the catalog stays organized and relevant instead of becoming another inbox to sort through.

SMART SEARCH THAT SURFACES THE RIGHT FIT

The system filters and ranks internships for each student based on their major and skills, and lets them search by company, deadline, or keyword. Relevant opportunities rise to the top in seconds, so students spend their time applying rather than hunting.

EFFORTLESS APPLICATIONS AND REAL-TIME TRACKING

Students apply through the system in a single step, and every application is recorded with its date and status, whether that is applied, interviewing, offer received, or rejected. Advisors see these updates live for the students they support, replacing one-by-one check-ins with a view that is always current.

REPORTING THAT PROVES RESULTS

The system generates reports for faculty and advisors, such as how many students have applied, interviewed, or received offers, broken down by major. These numbers help the program spot students who need help and demonstrate its impact to stakeholders.

BUILT TO BE DEPENDABLE

A system this central must be trustworthy, fast, and easy to use. The solution is engineered with that in mind.

- Secure by design. Access is based on each person's role. Students see and edit only their own profile and applications, while advisors and faculty have the added permissions, they need to manage listings and view progress. Sensitive information such as GPA is protected, and the system follows the university's data-handling standards.

- Fast when it matters. Searches return quickly, and the system supports many users at once, including the rush at the start of each semester.
- Reliable and available. The database is available throughout the day and backed up regularly so that information is never lost.
- Easy for everyone. The interface is simple enough for a first-time student user while still giving advisors efficient tools, and it works across the devices and browsers people already use.
- Ready to grow. The design accommodates new majors, skills, companies, and larger cohorts without rework, so the system keeps pace as the program grows.

HOW THE SOLUTION IS DESIGNED

Underneath the simple experience is a carefully structured database that makes the matching and tracking possible. The system is organized around ten tables: Student, Major, Skill, Advisor, Company, Internship, and Application, plus three connecting tables (StudentSkill, InternshipSkill, and Advises) that link records together where many items relate to many others. The Entity Relationship Diagram below shows these pieces and how they connect, and the Relational Schema that follows shows how they become the actual tables the system runs on.

ENTITY RELATIONSHIP DIAGRAM

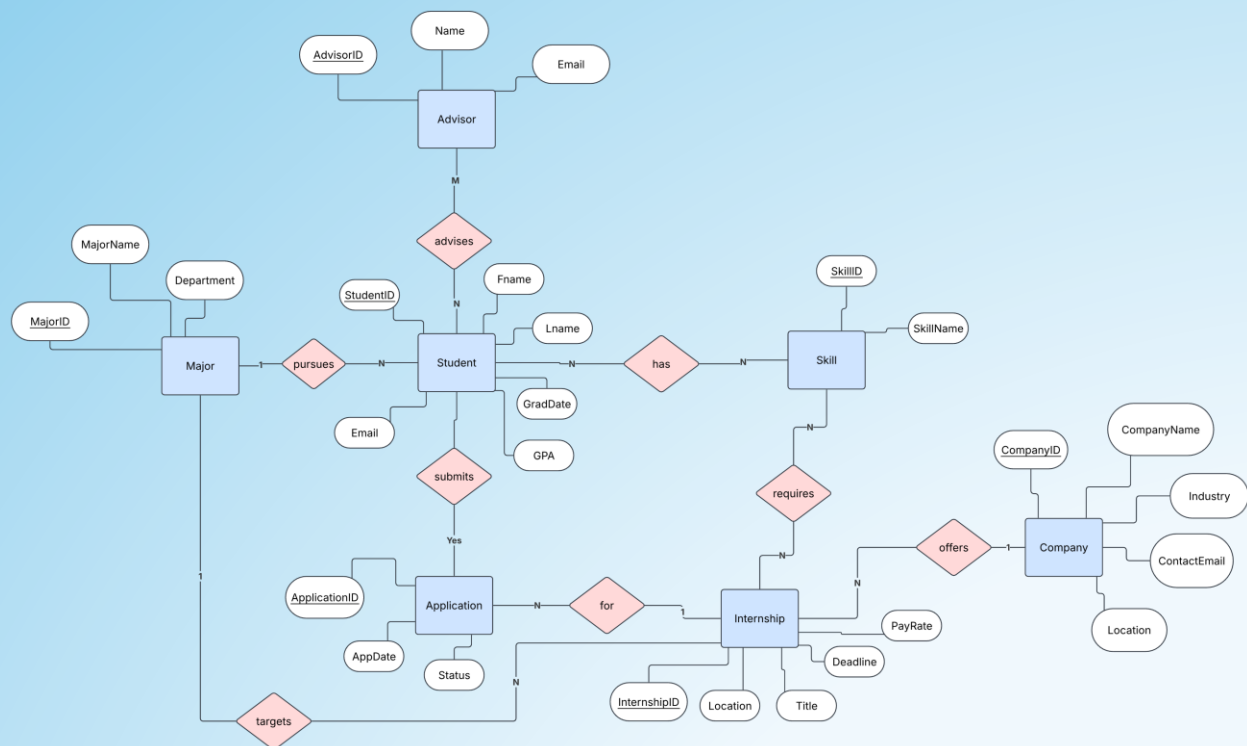


Figure 1. Entity Relationship Diagram for the Internship Matching Database.

RELATIONAL SCHEMA

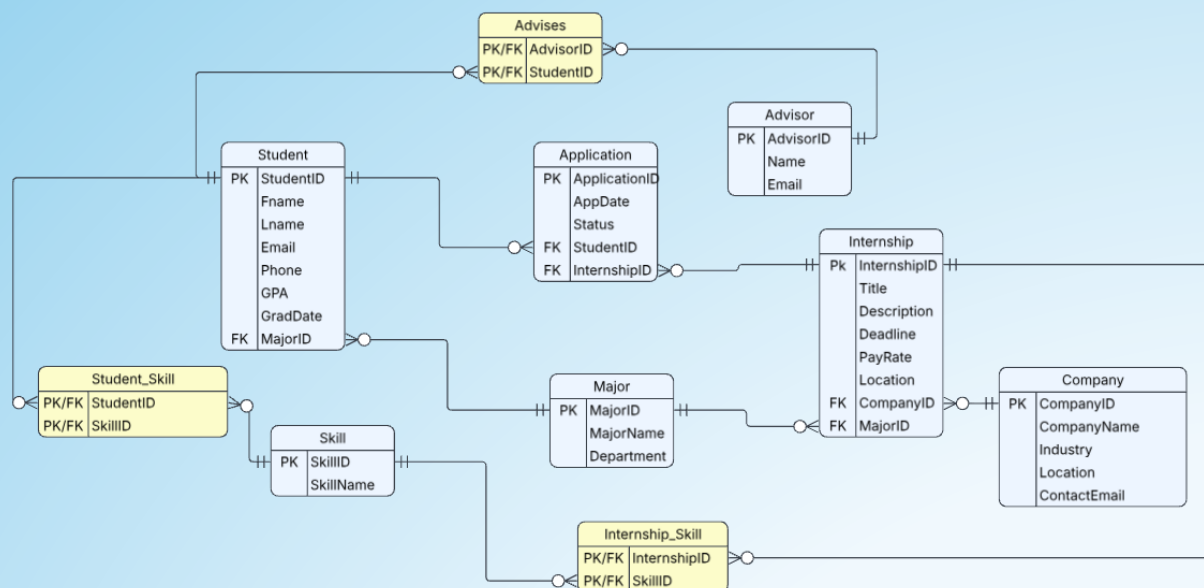


Figure 2. Relational Schema derived from the ERD.

DATA REFERENCE

For readers who want the detail, the tables below list every field the system stores and how the tables connect. PK marks the field that uniquely identifies each record, and FK marks a field that links to another table.

Student

Attribute	Key	Type	Description
StudentID	PK	INT	Unique identifier for each BUS 395 student.
FName		VARCHAR	Student first name.
LName		VARCHAR	Student last name.
Email		VARCHAR	Student contact email.
Phone		VARCHAR	Student phone number.
GPA		DECIMAL	Current grade point average.
GradDate		DATE	Expected graduation date.
MajorID	FK	INT	References Major(MajorID).

Major

Attribute	Key	Type	Description
MajorID	PK	INT	Unique identifier for each major.

Attribute	Key	Type	Description
MajorName		VARCHAR	Name of the major (e.g., MIS, Finance).
Department		VARCHAR	Owning Shidler department.

Skill

Attribute	Key	Type	Description
SkillID	PK	INT	Unique identifier for each skill.
SkillName		VARCHAR	Name of the skill (e.g., SQL, Excel).

StudentSkill (link table)

Attribute	Key	Type	Description
StudentID	PK, FK	INT	References Student (StudentID).
SkillID	PK, FK	INT	References Skill (SkillID).

Advisor

Attribute	Key	Type	Description
AdvisorID	PK	INT	Unique identifier for each advisor.
Name		VARCHAR	Advisor full name.
Email		VARCHAR	Advisor contact email.

Advises (link table)

Attribute	Key	Type	Description
AdvisorID	PK, FK	INT	References Advisor (AdvisorID).
StudentID	PK, FK	INT	References Student (StudentID).

Company

Attribute	Key	Type	Description
CompanyID	PK	INT	Unique identifier for each company.
CompanyName		VARCHAR	Name of the hosting company.
Industry		VARCHAR	Industry sector.
Location		VARCHAR	Company location.
ContactEmail		VARCHAR	Company point-of-contact email.

Internship

Attribute	Key	Type	Description
InternshipID	PK	INT	Unique identifier for each listing.
Title		VARCHAR	Internship title.
Description		TEXT	Description of the role.

Attribute	Key	Type	Description
Deadline		DATE	Application deadline.
PayRate		DECIMAL	Hourly or stipend pay rate.
Location		VARCHAR	Internship location.
CompanyID	FK	INT	References Company (CompanyID).
MajorID	FK	INT	References Major (MajorID). The target major for this listing.

InternshipSkill (link table)

Attribute	Key	Type	Description
InternshipID	PK, FK	INT	References Internship (InternshipID).
SkillID	PK, FK	INT	References Skill (SkillID).

Application

Attribute	Key	Type	Description
ApplicationID	PK	INT	Unique identifier for each application.
AppDate		DATE	Date the application was submitted.
Status		VARCHAR	Applied, Interviewing, Offer, or Rejected.

Attribute	Key	Type	Description
StudentID	FK	INT	References Student (StudentID).
InternshipID	FK	INT	References Internship (InternshipID).

WHY THE DESIGN WORKS

The structure was shaped directly by what BUS 395 needs. It centers on seven main records, Student, Major, Skill, Advisor, Company, Internship, and Application, with three small connecting tables that link them, where one item relates to many others. A few design choices are worth calling out because they are what make the matching and tracking possible.

Every student belongs to one major, and each major can have many students. Keeping majors in their own table lets the program organize and report on internships by major without repeating that information on every student record. This is the backbone of major-based matching.

Students and skills connect in both directions: a student can have many skills, and any given skill can belong to many students. The same is true for internships and the skills they require. The system handles these many-to-many links with small connecting tables, and it is exactly this design that lets the system compare a student's skills against the skills a listing calls for and surface the best fits.

Advisors and students are linked in the same way, since an advisor supports many students and a student may work with more than one advisor over time. Keeping advisors in a separate table lets the program control which advisor sees which students' progress.

Each internship is offered by one company, while a company can offer many internships, and each internship targets one major. These simple links keep listings tied to the right employer and the right audience, which keeps the catalog clean and relevant.

Applications sit at the center of the whole workflow. A student can submit many applications, and an internship can receive many, so each application records one student's submission to one internship, along with its date and status. This is the record advisors rely on to see briefly who has applied, who is interviewing, and who has an offer.

THE EXPERIENCE FOR STUDENTS AND ADVISORS

The design only matters if it is easy to use, so the system presents itself through three simple screens: a secure login, a student view for finding and applying to internships, and an advisor view for tracking progress.

SIGNING IN

Everyone signs in with their University of Hawai‘i credentials, and the system shows each person the right experience based on their role. Students go straight to their matched internships, while advisors and faculty arrive at the tools they need to manage listings and monitor progress. New students complete a short registration that sets up their profile.

THE STUDENT VIEW

After signing in, a student sees internships matched to their major and skills, each showing the title, company, location, pay rate, and deadline. They can search and filter listings, open any one for the full details, and apply with a single action. A simple “My Applications” view shows the status of everything they have applied to, so they always know where they stand.

THE ADVISOR VIEW

Advisors see a roster of the students they support alongside each student’s application activity. From this one dashboard, an advisor can post and maintain internship listings, update an application’s status, and generate reports that summarize how many students have applied, interviewed, or received offers in each major. What used to take a dozen separate conversations now happens immediately.

THE PAYOFF

With this system in place, the everyday experience of BUS 395 changes for everyone involved. Students get one reliable place to find internships that fit them, search and filter in seconds, apply in a single step, and track every application without digging through old emails and separate websites.

Advisors and faculty trade one-by-one status chasing for a single, always-current view. They can post and maintain listings, connect employers to opportunities, see exactly where each student stands, and pull reports that show how the program is performing by major. That visibility makes it far easier to step in early when a student needs help.

Best of all, the matching the program has always wanted happens automatically. Because internships are organized by major and required skills, relevant suggestions surface on their own instead of relying on someone to make the connection by hand. The program spends less effort coordinating and gets better results to show for it.

MOVING FORWARD

BUS 395 has a clear problem and a clear opportunity. Internship information is scattered, students miss good matches, and advisors have no easy way to see the whole picture. The Internship Matching Database solves all three by bringing everything into one place, matching students to the right opportunities, and giving advisors real-time visibility into progress.

From here, the path is straightforward. We will build the database from the design shown in this proposal, load the initial information on majors, skills, companies, and current listings, and test the system thoroughly against the goals described here. Once it is validated, we will roll it out within the university environment and keep it maintained as new cohorts arrive and the program grows. We are

confident this solution will save time, improve outcomes, and give the Shidler College of Business a lasting advantage in connecting its students with the right internships, and we look forward to building it.

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